Breaking Apart Wet Weather Flow Signals Allows Streamlined Auto-calibration of Sewer Hydrology Models

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Abstract

Modeling the impacts of precipitation on sanitary sewer flow is challenging due to limited-duration flow metering records, complex interactions among static and time-varying factors, and short-term, long-term, and seasonal hydrologic processes. The traditional calibration approach adjusts and optimizes parameters for all processes simultaneously to attempt to match the measured flow rate.

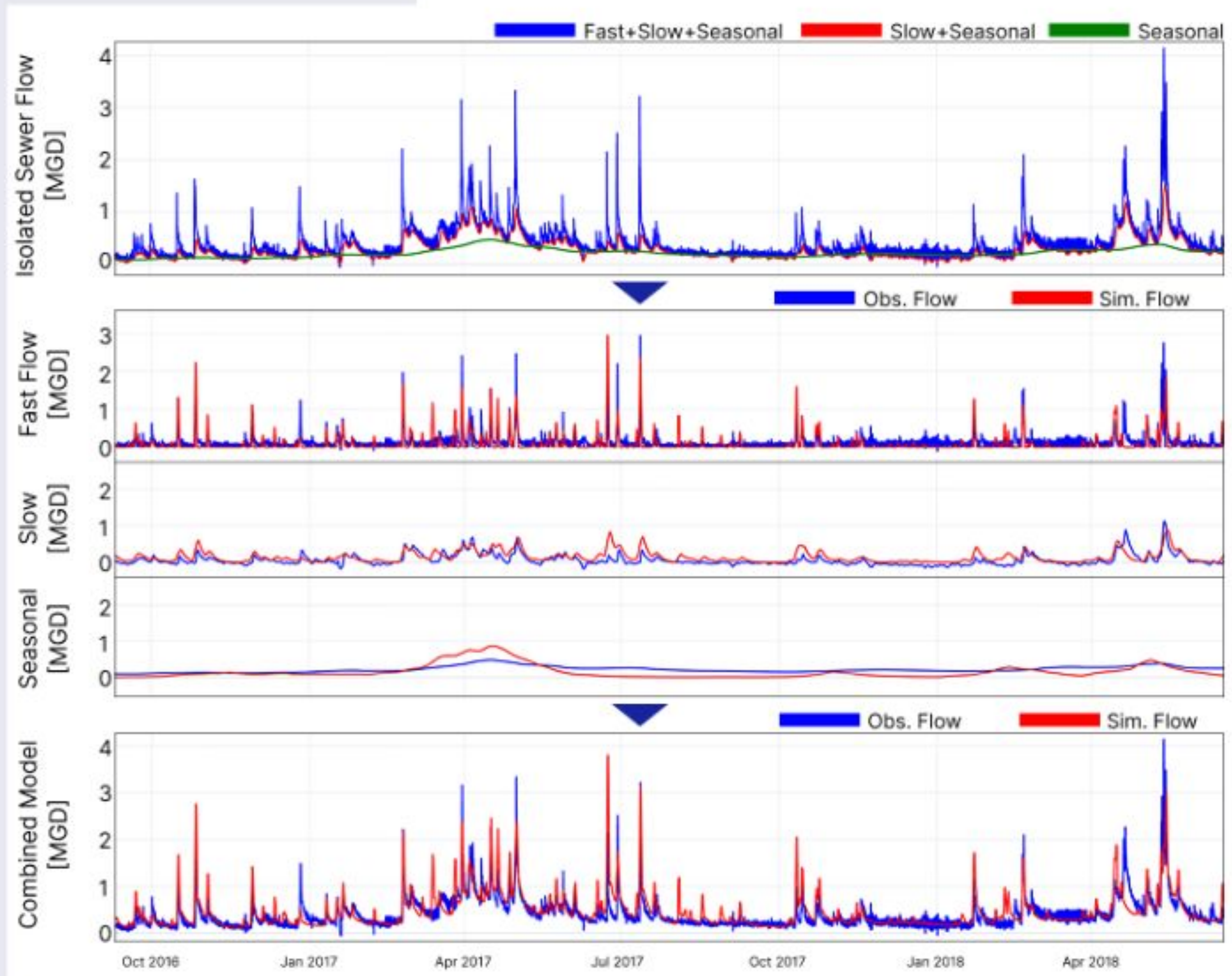
In this study, we present an alternative calibration approach that first disaggregates the measured sewer flow signal into four sub-signals, each of which represents a different hydrologic process. Then each hydrologic process is independently modeled and calibrated to its own distinct sub-signal. This approach significantly reduces the parameter solution domain, resulting in faster, and more reliable and confident calibration.

Objective

In this study, we compare the calibration effort and predictive accuracy of the traditional calibration approach of sewer hydrology modeling to the alternative approach of disaggregating the total flow signal into distinct sub-signals. We aim to quantify an anticipated improvement in required model calibration effort and reliability of model parameter estimates.

Methods

MS0208 Sewer, Observed and Simulated Flow



In disaggregating the total flow signal, we focused on four distinct hydrologic processes: diurnal variation, long-term seasonal variation, slow response rainfall derived infiltration and inflow (RDI/I), and fast response RDI/I. First, we removed the diurnal variation from the total flow using pattern recognition techniques. From there, we used base-flow isolation filters to identify and isolate the long-term seasonal variation and the slow RDI/I. The remaining portion of the signal was the fast RDI/I signal. For base flow isolation, we used Lynn-Hollick (1979) filter with an alpha of 0.925. We used five passes (three forward and two backward) on daily flow values to isolate the seasonal variation. After removing the seasonal variation, we used five passes on hourly flow values to isolate slow RDI/I.

We assumed the diurnal variation stays constant throughout the training and modeling periods for both the traditional and disaggregated calibration approaches. The remaining sub-signals were modeled using the Antecedent Moisture Model (AMM) approach as described by Edgren, et.al (2024). For seasonal variation modeling, we used a four-parameter AMM. We used two six-parameter AMMs for the slow and fast RDI/I. The traditional calibration approach involved calibrating all sixteen parameters simultaneously. The disaggregated calibration approach involved calibrating the four or six sub-signal model parameters independently.

We used Python to run a parameter optimization algorithm based on a weighted objective function. Using Scikit Learn as our main machine learning library, we developed separate modules that streamline the process of signal disaggregation, model calibration, model prediction, and results presentation.

For this specific case study, we used hourly flow data provided by the Milwaukee Metropolitan Sewerage District.

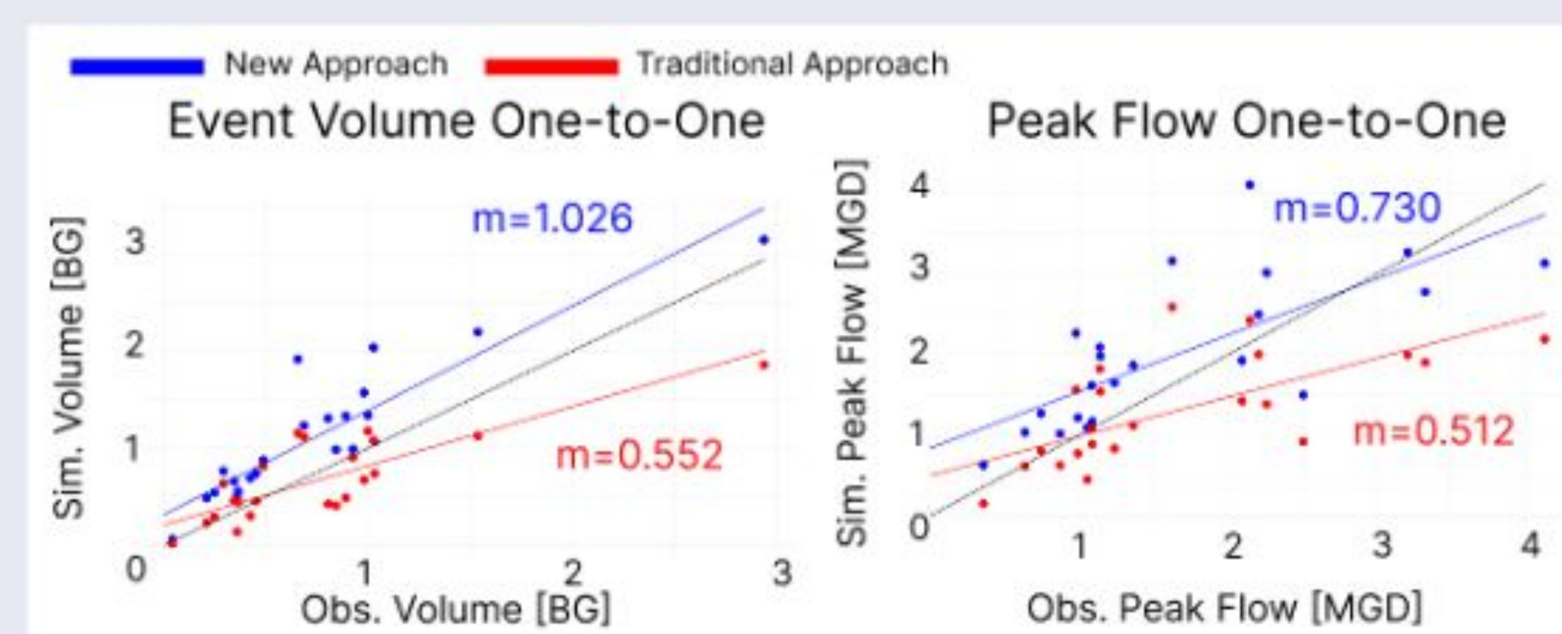
Analysis & Result

The disaggregated calibration optimization convergence required an average of 35–45% fewer iterations compared to the traditional calibration approach. The mean calibration runtime decreased by approximately 25%.

Across all validation periods, the disaggregated calibration approach achieved lower error metrics than the traditional approach. Root Mean Square Error (RMSE) decreased by 15-20%.

Average peak flow prediction error decreased by 30%, and event volume prediction error decreased by 47%. Parameter variability across repeated optimization runs was also reduced.

When independently modeled sub-signals were combined, the modeled peak flow rate and event volume showed less bias than values predicted by the models calibrated with the traditional approach. Overall, the disaggregated framework demonstrated measurable improvements in calibration efficiency, predictive accuracy, and parameter robustness relative to the traditional combined-model approach.



Conclusion & Discussion

Sanitary sewer flow signal disaggregation into diurnal variation, seasonal variation, slow RDI/I, and fast RDI/I sub-signals successfully created the situation where each process could be independently calibrated and modeled with a subset of total model parameters. Independent modeling reduced calibration complexity and significantly improved optimization efficiency, consistent with the curse of dimensionality. The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings (Wikipedia). The disaggregated signal calibration approach reduces model dimensions three-fold which makes it exponentially easier to calibrate than the traditional calibration approach.

The disaggregated calibration approach achieved improved predictive accuracy relative to a traditional calibration approach. Although not quantified, it is suspected that the joint confidence region of the optimal parameter set is much smaller and consequently more robust, too.

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Explore Project Results

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Edgren, D., Czachorski, R., Gonwa, W. 2024 “Reparameterizing the Antecedent Moisture Model”, Journal of Water Management Modeling. RJN Group, Inc., OHM Advisors, H20metrics, Milwaukee School of Engineering. https://*doi.org/10.14796/JWMM.C525*

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